WHAT ARE KEY ADAS FEATURES?

Depending on the level of functionality and tech offered, you can expect the following features in a modern car with ADAS



Adaptive Cruise Control (ACC)

Just like regular cruise control, only intelligent. Upon setting the ACC control on, the system uses the front cameras and sensors to gauge the distance from the vehicle ahead and accordingly adjust the speed. ACC always maintains a safe braking distance from the vehicle in front.

Automatic Emergency Braking (AEB)

In conjunction with ACC or independently, AEB Uses its array of sensors and cameras to determine the speed of the vehicle in front and apply brakes if the minimum safe braking distance is suddenly shortened.

Forward Collision Warning (FCW)

In Level 0-enabled ADAS cars, the system detects close proximity to the vehicle or object in front and warns the driver, who then takes the evasive action. A Level 1 or 2-equipped system uses AEB to maximise braking to avoid the collision or minimise the damage.

Lane Departure Warning (LDW)

Veering off a lane is the first sign of a driver's lack of attention. LDW systems will issue an audio warning, or also a vibration on the steering wheel, to alert the driver.

Blind Spot Detection (BSD)

The side/ORVM-mounted cameras have a wider field of vision to detect vehicles in the blind spot of the door mirrors. Even with convex blind spot mirrors, there is still a possibility of missing smaller objects like motorcyclists.

Automatic Parking Assistance

Once only restricted to high-end cars, this system is now seen on many cars. The system gauges the space (both parallel or vertical) to park the vehicle and completely takes over steering control with only braking left to the driver's discretion.





Other functions

A few other intelligent functions are also included in ADAS like adaptive headlamps that auto-dim from high beam on recognising oncoming traffic or switch it on to illuminate an empty, unlit road. Drowsiness detection with cameras to determine falling level of attention based on eye movements and similar warning functions are available.

there is a need for standardisation so they can work effectively in a variety of conditions. Level 3 to Level 5 systems offer advanced to fully autonomous driving, where human intervention is minimal or not required at all and the decision-making rests upon the system within preset parameters, or even AI-enabled actions. These features are currently under development and some of them have not yet made it to widespread commercial use.

Can ADAS be used in all conditions?

The gradual adoption of ADAS by manufacturers and offering them at more affordable price points will surely see a decrease in road accidents and fatalities in the coming years. The bigger question is trusting the system. There is a learning curve, albeit a short one, into understanding how and in what scenarios the system works best. India presents unique traffic scenarios that vary from organised to chaotic, from completely built to lack of road infrastructure, and many variables as a result of the lack of driving discipline and vagaries of law enforcement.

In our tests, we found it best to use ADAS on roads with clear lane demarcation, predictable and moderate traffic. If the variables on the road are reduced, the set parameters do a fine job of reducing driving stress while offering some level of autonomous driving. The current level of ADAS available in cars in India always demands the driver's attention with their hands on the steering wheel at all times. This is a great start in developing trust in the automated system and technology as we build our reliance on these systems. ■